

B. VINETHA PALLAVI

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EDUCATION

SRM University

Andhra Pradesh, India

- Cumulative GPA: 7.18 / 10

B.Tech. Electronics and Communications Engineering

2024 – 2028

PROJECTS

Real-Time Object Detection Using MATLAB

- Developed a real-time object detection system using image processing techniques in MATLAB.
- Captured live video using a webcam and processed individual frames for object recognition.
- Applied image filtering methods to improve detection accuracy and performance.

Closed Loop Control System Analysis using Simulink

- Designed and analyzed a closed-loop control system using MATLAB Simulink.
- Modeled system response to study performance metrics including stability, overshoot, and settling time.
- Compared open-loop vs closed-loop responses through graphical analysis.
- Enhanced system performance by fine-tuning controller parameters.

Reinforcement Learning-Based Accident Prediction System

- Implemented Deep Q-Network (DQN) using MATLAB Reinforcement Learning Toolbox.
- Used a traffic accident dataset with 12,000+ records and 32 features for state modeling.
- Designed a reward-based system to support safe driving decision-making.
- Achieved 89% accuracy and an F1-score of 0.94 on the prediction task.
- Visualized training performance using episode reward graphs.
- Focused on real-time accident prevention rather than post-event prediction.

CERTIFICATIONS

- MATLAB Onramp — MathWorks
- Simulink Onramp — MathWorks
- Control Design Onramp with Simulink — MathWorks

SKILLS

Programming Languages: C, Python

Tools and Technologies: MATLAB, Simulink, MATLAB Reinforcement Learning Toolbox, Deep Q-Network (DQN)

Technical Skills: Machine Learning, Reinforcement Learning, Data Analysis, Image Processing, Control Systems, Object Detection

Soft Skills: Teamwork, Problem Solving, Communication

Languages Known: English, Telugu